

1 Functions f and g are defined on the set of real numbers by

- $f(x) = x^2 + 3$
- $g(x) = x + 4$

(a) Find expressions for:

- (i) $f(g(x))$;
- (ii) $g(f(x))$.

3

Generic Scheme

Illustrative Scheme

1 (a)

- ¹ ic start composite process
- ² ic correct substitution into expression
- ³ ic complete second composite

- ¹ e.g. $f(x+4)$ stated, or implied by •²
- ² $(x+4)^2 + 3$
- ³ $x^2 + 3 + 4$

3

Notes

- Candidates must clearly identify which of their answers are $f(g(x))$ and $g(f(x))$; the minimum evidence for this could be as little as using (i) and (ii) as labels.
- Candidates who interpret the composite functions as either $f(x) \times g(x)$ or $f(x) + g(x)$, do not gain any marks.

Regularly occurring responses

Response 1 : The first two marks are for **either** $f(g(x))$ **or** $g(f(x))$ correct. The third mark is for the other composite function.

Candidate A

$$f(g(x)) \\ = (x+4)^2 + 3 \quad \checkmark \bullet^1 \checkmark \bullet^2$$

$$g(f(x)) \\ = x^2 + 12 \quad \times \bullet^3$$

2 marks out of 3

Candidate B

$$f(g(x)) \\ = (x+7)^2 \quad \times \bullet^3$$

$$g(f(x)) \\ = x^2 + 7 \quad \checkmark \bullet^1 \checkmark \bullet^2$$

2 marks out of 3

Response 2 : Interpreting $f(g(x))$ as $g(f(x))$ and vice versa. A maximum of 2 marks are available.

Candidate C

$$f(g(x)) \\ = x^2 + 7 \quad \times \bullet^1 \times \bullet^2$$

$$g(f(x)) \\ = (x+4)^2 + 3 \quad \checkmark \bullet^3$$

2 marks out of 3

Candidate D

$$f(g(x)) \\ = x^2 + 7 \quad \times \bullet^1 \times \bullet^2$$

1 mark out of 3

Response 3 : Identifying $f(g(x))$ and $g(f(x))$

Candidate E

$$(x+4)^2 + 3 \quad \times \bullet^1 \checkmark \bullet^2 \\ x^2 + 7 \quad \checkmark \bullet^3$$

2 marks out of 3

Candidate F

$$x^2 + 7 \quad \times \bullet^1 \times \bullet^2 \\ (x+4)^2 + 3 \quad \checkmark \bullet^3$$

1 mark out of 3

Candidate G

$$x^2 + 7 \quad \text{ONLY} \\ \text{or } (x+4)^2 + 3 \quad \text{ONLY}$$

0 marks out of 3

Candidate H

$$(i) (x+4)^2 + 3 \quad \checkmark \bullet^1 \checkmark \bullet^2 \\ (ii) x^2 + 7 \quad \checkmark \bullet^3$$

3 marks out of 3

Generic Scheme

Illustrative Scheme

1 (b)

Method 1 : Discriminant

- ⁴ pd obtain a quadratic expression
- ⁵ ss know to and use discriminant
- ⁶ ic interpret result

Method 2 : Quadratic Formula

- ⁴ pd obtain a quadratic expression
- ⁵ ss know to and use quadratic formula
- ⁶ ic interpret result

Method 1 : Discriminant

- ⁴ $2x^2 + 8x + 26$
- ⁵ $8^2 - 4 \times 2 \times 26$ or $4^2 - 4 \times 1 \times 13$ stated, or implied by •⁶
- ⁶ $-144 < 0$ or $-36 < 0$ so no real roots

Method 2 : Quadratic Formula

- ⁴ $2x^2 + 8x + 26$
- ⁵ $\frac{-8 \pm \sqrt{8^2 - 4 \times 2 \times 26}}{2 \times 2}$ stated, or implied by •⁶
- ⁶ $\sqrt{-144}$ not possible so no real roots

3

Notes

3. Candidates who use $f(x) \times g(x)$ can gain no marks in (b) as a cubic will be obtained.
4. Candidates who use $f(x) + g(x)$ do not gain •⁴ (eased) but •⁵ and •⁶ are available as follow through marks.
5. In method 1, any other formula masquerading as a discriminant cannot gain •⁵ and •⁶.
6. •⁴, •⁵ and •⁶ are only available if $f(g(x)) + g(f(x))$ simplifies to a quadratic expression of the form $ax^2 + bx + c$, with b and c both non-zero.
7. •⁶ is only available for a numerical value, calculated correctly from the candidate's response at •⁴, and leading to no real roots.
8. Do not accept for •⁶:
 - 'no roots' in lieu of 'no real roots'
 - 'maths error' or 'ma error'.
9. Candidates who use the word derivative instead of discriminant should not be penalised.

Regularly occurring responses

Response 4 : Candidates who do not simplify the value of their discriminant**Candidate I**

$$8^2 - 4 \times 2 \times 26 \quad \checkmark \quad \bullet^5 \quad \checkmark$$

$$= 64 - 208 < 0 \text{ so no real roots} \quad \bullet^6 \quad \times$$

Response 5 : Acceptable communication marks**Method 1****Candidate J**

$$\sqrt{8^2 - 4 \times 2 \times 26} \quad \checkmark \quad \bullet^5$$

$$= \sqrt{-144}$$

not valid

so no real roots $\checkmark \quad \bullet^6$

Candidate L

no real roots if $b^2 - 4ac < 0$

$$64 - 208 = -144 \quad \checkmark \quad \bullet^6$$

Candidate K

Discriminant $= \sqrt{8^2 - 4 \times 2 \times 26} \quad \checkmark \quad \bullet^5$

$$= \sqrt{-144}$$

can't find root of negative

so no real roots $\checkmark \quad \bullet^6$

Method 2**Candidate M**

$$\frac{-(-4) \pm \sqrt{8^2 - 4 \times 2 \times 26}}{2 \times 2} \quad \checkmark \quad \bullet^5$$

$$= \frac{4 \pm \sqrt{-144}}{4}$$

no $\sqrt{-ve}$

so no real roots $\checkmark \quad \bullet^6$

- 2 (a) Relative to a suitable set of coordinate axes, diagram 1 shows the line $2x - y + 5 = 0$ intersecting the circle $x^2 + y^2 - 6x - 2y - 30 = 0$ at the points P and Q.
- Find the coordinates of P and Q.

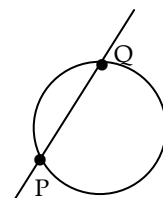


Diagram 1

6

Generic Scheme

Illustrative Scheme

2 (a)

- ¹ ss rearrange linear equation
- ² ss substitute into circle
- ³ pd express in standard form
- ⁴ pd start to solve
- ⁵ ic state roots
- ⁶ pd determine corresponding y -coordinates

Substituting for y

- ¹ $y = 2x + 5$ **stated, or implied by** •²
- ² $\dots (2x + 5)^2 \dots - 2(2x + 5) \dots$
- ³ $5x^2 + 10x - 15$ } = 0 must appear at the •³
- ⁴ e.g. $5(x + 3)(x - 1)$ } or •⁴ stage to gain •³.
- ⁵ $x = -3$ and $x = 1$
- ⁶ $y = -1$ and $y = 7$

Substituting for x

- ¹ $x = \frac{y - 5}{2}$ **stated, or implied by** •²
- ² $\left(\frac{y - 5}{2}\right)^2 \dots - 6\left(\frac{y - 5}{2}\right) \dots$
- ³ $5y^2 - 30y - 35$ } = 0 must appear at the •³
- ⁴ e.g. $5(y + 1)(y - 7)$ } or •⁴ stage to gain •³.
- ⁵ $y = -1$ and $y = 7$
- ⁶ $x = -3$ and $x = 1$

6

Notes

- At •⁴ the quadratic must lead to two real distinct roots for •⁵ and •⁶ to be available.
- Cross marking is available here for •⁵ and •⁶.
- Candidates do not need to distinguish between points P and Q.

Regularly occurring responses

Response 1 : Solving quadratic equation

Candidate A

✓ •¹ ✓ •²

$$5x^2 + 10x + 5 = 0 \quad \times \bullet^3$$

$$5(x + 1)(x + 1) \quad \times \bullet^4$$

$$x = -1 \quad \times \bullet^5$$

$$y = 3 \quad \times \bullet^6$$

Candidate B

$y = 2x + 5$ ✓ •¹

$$x^2 + (2x + 5)^2 - 6x - 2(7x + 5) - 30 = 0 \quad \times \bullet^2$$

$$5x^2 - 15 = 0 \quad \times \bullet^3$$

$$x^2 = 3 \quad \times \bullet^4$$

$$x = \pm\sqrt{3} \quad \times \bullet^5$$

$$y = 8.5, 1.5 \quad \times \bullet^6$$

Candidate C

✓ •¹ ✓ •²

$$5x^2 + 10x - 15 = 0 \quad \checkmark \bullet^3$$

$$5x^2 + 10x = 15$$

$$5x(x + 2) = 15 \quad \times \bullet^4$$

$$x(x + 2) = 3$$

$$x = 3 \quad x = 1 \quad \times \bullet^5$$

$$y = 11 \quad y = 7 \quad \times \bullet^6$$

Cross marking is **not** available here for •⁵ and •⁶, as there are no distinct roots. See Note 1.

- 2 (b) Diagram 2 shows the circle from (a) and a second congruent circle, which also passes through P and Q.
Determine the equation of this second circle.

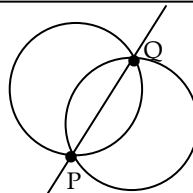


Diagram 2

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Generic Scheme

Illustrative Scheme

2 (b)

- ⁷ ic centre of original circle
- ⁸ pd radius of original circle

Method 1 : Using midpoint

- ⁹ ss midpoint of chord
- ¹⁰ ss evidence for finding new centre
- ¹¹ ic centre of new circle
- ¹² ic equation of new circle

Method 2 : Stepping out using P and Q

- ⁹ ss evidence of C_1 to P or C_1 to Q
- ¹⁰ ss evidence of Q to C_2 or P to C_2
- ¹¹ ic centre of new circle
- ¹² ic equation of new circle

- ⁷ (3, 1)

- ⁸ $\sqrt{40}$ Accept $r^2 = 40$

Method 1 : Using midpoint

- ⁹ (-1, 3)
- ¹⁰ e.g. stepping out or midpoint formula
- ¹¹ (-5, 5)
- ¹² $(x+5)^2 + (y-5)^2 = 40$

Method 2 : Stepping out using P and Q

- ⁹ e.g. stepping out or vector approach
- ¹⁰ e.g. stepping out or vector approach
- ¹¹ (-5, 5)
- ¹² $(x+5)^2 + (y-5)^2 = 40$

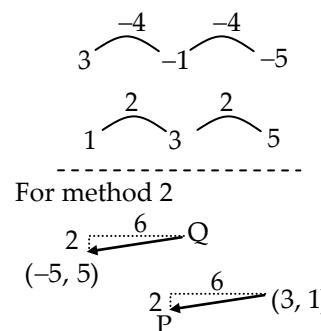
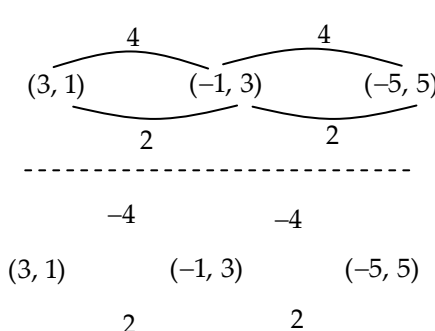
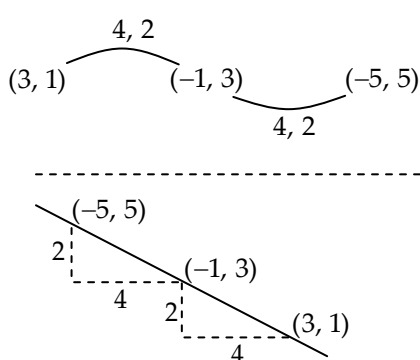
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Notes

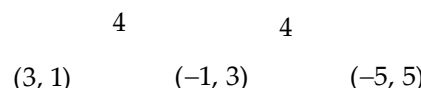
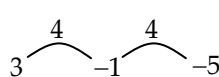
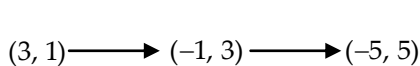
4. The evidence for •⁷ and •⁸ may appear in (a).
5. Centre (-5, 5) **without working** in method 1 may still gain •¹² but not •¹⁰ or •¹¹, in method 2 may still gain •¹² but not •⁹, •¹⁰ or •¹¹.
Any other centre **without working** in method 1 does not gain •¹⁰, •¹¹ or •¹², in method 2 does not gain •⁹, •¹⁰, •¹¹ or •¹².
6. The centre must have been clearly indicated before it is used at the •¹² stage.
7. Do not accept e.g. $\sqrt{40}^2$ or 39.69 , or any other decimal approximations for •¹².
8. The evidence for •⁸ may not appear until the candidate states the radius or equation of the second circle.

Regularly occurring responses

Response 2 : Examples of evidence for stepping out for •¹⁰ in method 1 or •⁹ or •¹⁰ in method 2



Response 3 : Examples of evidence which do not gain •¹⁰ in method 1 for stepping out



3 A function f is defined on the domain $0 \leq x \leq 3$ by $f(x) = x^3 - 2x^2 - 4x + 6$.

Determine the maximum and minimum values of f .

7

Generic Scheme

Illustrative Scheme

3

- ¹ ss start to differentiate
- ² ss complete derivative and set to 0
- ³ pd start to solve $f'(x) = 0$
- ⁴ pd solve $f'(x) = 0$
- ⁵ ic evaluate f at relevant stationary point
- ⁶ ss consider end-points
- ⁷ ic state max. and min. values

- ¹ differentiate x^3 or $-2x^2$ correctly
- ² $3x^2 - 4x - 4$
- ³ e.g. $(3x+2)(x-2)$ } = 0 must appear at •²
- ⁴ $-\frac{2}{3}, 2$
- ⁵ $f(2) = -2$
- ⁶ $f(0) = 6$ and $f(3) = 3$
- ⁷ max. 6 and min. -2

7

Notes

- The only valid approach is via differentiation. A numerical approach can only gain •⁶.
- Candidates who consider stationary points only cannot gain •⁶ or •⁷.
- Treat maximum (0, 6) and minimum (2, -2) as bad form.
- Cross marking is **not** applicable to •⁶ or •⁷.
- Ignore any nature table which may appear in a candidate's solution, however (2, -2) at table is sufficient for •⁵.

Regularly occurring responses

Response 1 : Algebraic issues in working

Candidate A

$$\begin{aligned}
 y' &= 3x^2 - 4x - 4 \quad \checkmark \\
 (3x-2)(x+2) &\quad \times \\
 x &= \frac{2}{3}, \quad x = -2 \quad \checkmark \\
 \text{When } x &= \frac{2}{3}, y = \frac{74}{27} \quad \checkmark \\
 f(0) &= 6 \text{ and } f(3) = 3 \quad \checkmark \\
 \text{max} &= 6, \quad \text{min} = 2\frac{20}{27} \quad \checkmark
 \end{aligned}$$

- ¹ ✓
- ² ✗
- ³ ✗
- ⁴ ✗
- ⁵ ✗
- ⁶ ✓
- ⁷ ✓

Candidate B

$$\begin{aligned}
 3x^2 - 4x - 4 &= 0 \quad \checkmark \\
 (3x-2)(x-2) &\quad \times \\
 x &= \frac{2}{3} \text{ or } x = 2 \quad \checkmark \\
 \text{so } f(2) &= -2 \quad \times
 \end{aligned}$$

- ¹ ✓
- ² ✓
- ³ ✗
- ⁴ ✗
- ⁵ ✗

Since $\frac{2}{3}$ is within the domain, $f\left(\frac{2}{3}\right)$ must also be calculated to gain •⁵.

Ignore the value of $f\left(-\frac{2}{3}\right)$ here, if it is included.

Candidate C

$$\begin{aligned}
 3x^2 - 4x - 4 &\quad \checkmark \\
 (3x+2)(x-2) &\quad \checkmark \\
 3x+2 &= 0 \quad x-2 = 0 \\
 x &= -\frac{2}{3} \quad x = 2 \quad \checkmark \\
 f(2) &= -2 \quad \checkmark
 \end{aligned}$$

- ¹ ✓
- ² ✗
- ³ ✓
- ⁴ ✓
- ⁵ ✓

Response 2 : Derivative not explicitly set to zero

Candidate D

$$\begin{aligned}
 f'(x) &= 3x^2 - 4x - 4 \quad \checkmark \\
 f'(x) &= 0 \quad \checkmark
 \end{aligned}$$

Candidate E

$$\begin{aligned}
 f'(x) &= 0 \\
 f'(x) &= 3x^2 - 4x - 4 \quad \checkmark \\
 &= (3x+2)(x-2) \quad \checkmark
 \end{aligned}$$

Candidate F

$$\begin{aligned}
 f'(x) &= 0 \\
 3x^2 - 4x - 4 &\quad \checkmark \\
 &= (3x+2)(x-2) \quad \checkmark
 \end{aligned}$$

Candidate G

$$\begin{aligned}
 f'(x) &= 0 \text{ only} \quad \times \\
 &\quad \times
 \end{aligned}$$

Regularly occurring responses

Response 3 : Solving quadratic equation

Candidate H

$$\begin{aligned}
 f'(x) &= 3x^2 - 4x - 4 & \bullet^1 \checkmark \\
 3x^2 - 4x - 4 &= 0 & \bullet^2 \checkmark \\
 3x^2 - 4x &= 4 & \bullet^3 \times \\
 x(3x - 4) &= 4 & \bullet^4 \times \\
 x = 4, \frac{4}{3} & & \bullet^5 \times
 \end{aligned}$$

Candidate I

$$\begin{aligned}
 3x^2 - 4x - 4 &= 0 & \bullet^1 \checkmark \\
 x &= \frac{-(-4) \pm \sqrt{(4)^2 - 4 \times 3 \times (-4)}}{2 \times 3} & \bullet^2 \checkmark, \bullet^3 \checkmark
 \end{aligned}$$

Ignore omission of negative sign at square here.

Due to 'method' chosen $\bullet^3$, $\bullet^4$, $\bullet^5$ and $\bullet^7$ are not available.

Response 4 : Numerical approach

Candidate J

$$\begin{aligned}
 f(0) &= 6 \\
 f(3) &= 3 & \bullet^6 \checkmark
 \end{aligned}$$

This candidate has stayed within the interval $0 \leq x \leq 3$.

Candidate K

$$\begin{aligned}
 f(0) &= 6 \\
 f(1) &= 1 \\
 f(2) &= -2 & \bullet^5 \times \\
 f(3) &= 3 & \bullet^6 \checkmark
 \end{aligned}$$

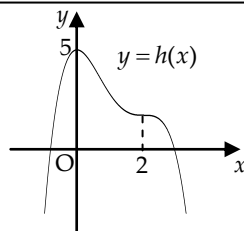
Candidate L

$$\begin{aligned}
 f(0) &= 6 \\
 f(1) &= 1 \\
 f(2) &= -2 & \bullet^5 \times \\
 f(3) &= 3 & \bullet^6 \times \\
 f(4) &= 22
 \end{aligned}$$

This candidate has gone outwith the interval $0 \leq x \leq 3$.

For $\bullet^5$, $f(2)$ must come from calculus and not from any other approach.

- 4 The diagram below shows the graph of a quartic $y = h(x)$, with stationary points at $x = 0$ and $x = 2$.



On separate diagrams sketch the graphs of:

- (a) $y = h'(x)$;
(b) $y = 2 - h'(x)$.

3

3

Generic Scheme

Illustrative Scheme

4 (a)

- ¹ ic identify roots
- ² ic interpret point of inflection
- ³ ic complete cubic curve

- ¹ 0 and 2 only
- ² turning point at (2, 0)
- ³ cubic, passing through O with negative gradient

3

Notes

1. All graphs must include both the x and y axes (labelled or unlabelled), however the origin need not be labelled.
2. No marks are available unless a graph is attempted.
3. No marks are available to a candidate who makes several attempts at a graph on the same diagram, unless it is clear which is the final graph.
4. A linear graph gains no marks in both (a) and (b).

4 (b)

- ⁴ ic reflection in x -axis
- ⁵ ic translation $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$
- ⁶ ic annotation of 'transformed' graph

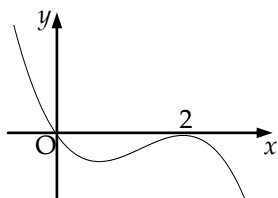
- ⁴ reflection of graph in (a) in x -axis
- ⁵ graph moves parallel to y -axis by 2 units upwards
- ⁶ two 'transformed' points appropriately annotated (see Note 5)

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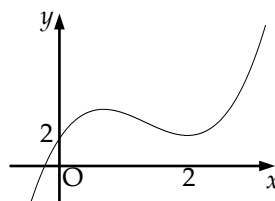
Notes

5. 'Transformed' here means a reflection followed by a translation.
6. •⁴ and •⁵ apply to the entire curve.
7. In each of the following circumstances :
 - Candidates who transform the original graph
 - Candidates who sketch a parabola in (a)
 mark the candidate's attempt as normal and unless a mark of 0 has been scored, deduct the last mark awarded. Indicate this with ✖ (see Regular occurring response G).
8. A reflection in any line parallel to the y -axis does not gain •⁴ or •⁶.
9. A translation other than $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$ does not gain •⁵ or •⁶.

Graph for (a)



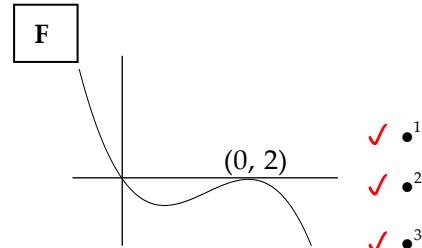
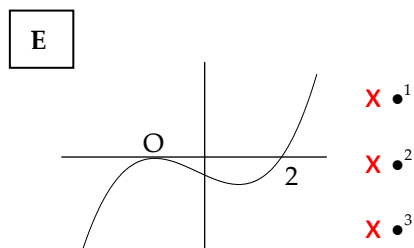
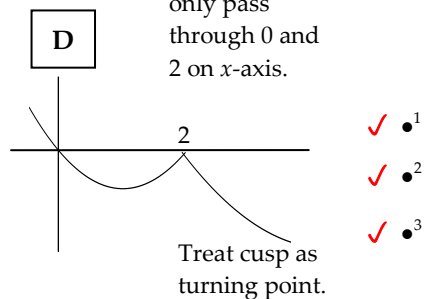
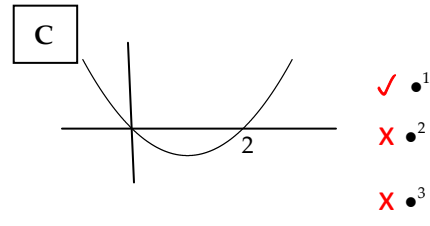
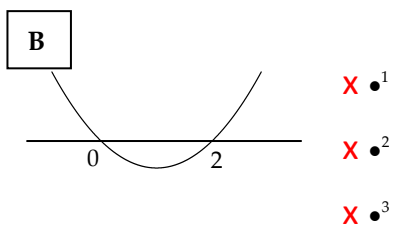
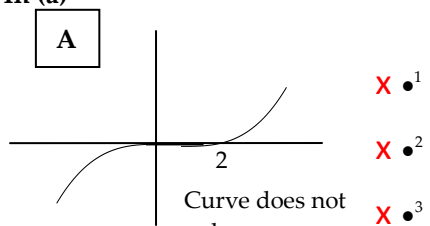
Graph for (b)



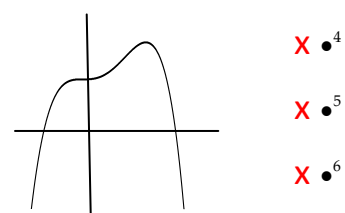
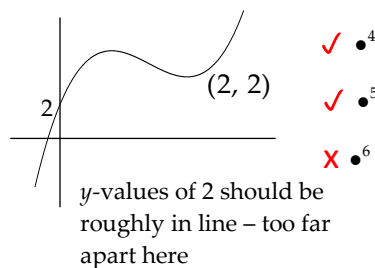
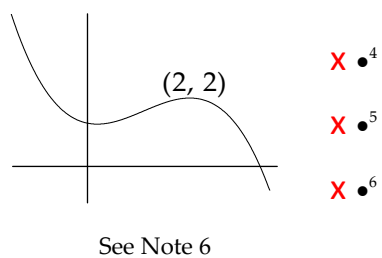
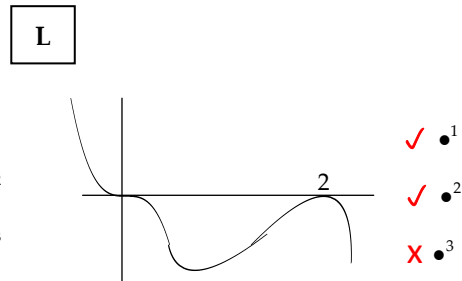
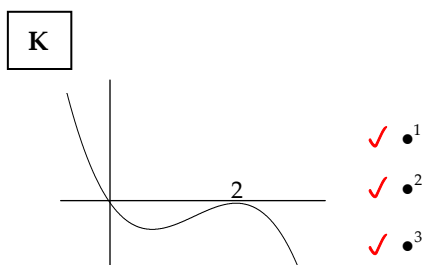
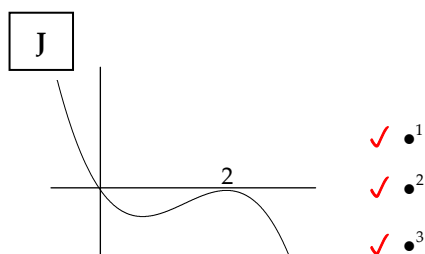
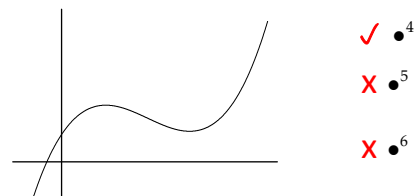
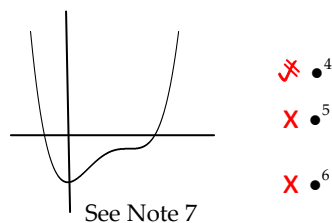
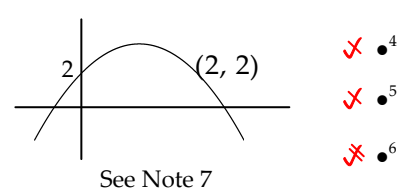
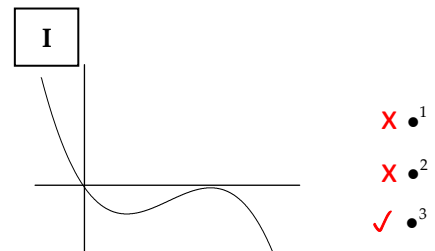
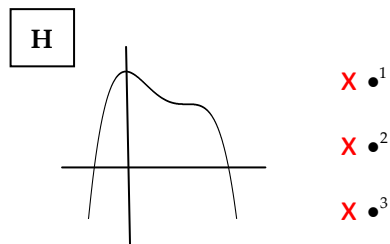
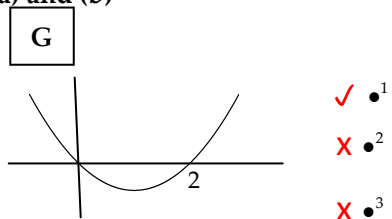
- ⁴ ic reflection in x -axis

Regularly occurring responses

In (a)



In (a) and (b)



5 A is the point (3, -3, 0), B is (2, -3, 1) and C is (4, k, 0).

(a) (i) Express $\overrightarrow{BA}$ and $\overrightarrow{BC}$ in component form.

(ii) Show that $\cos \hat{ABC} = \frac{3}{\sqrt{2(k^2 + 6k + 14)}}$.

7

Generic Scheme

Illustrative Scheme

5(a)

•¹ ic interpret vector

•² pd process vector

•³ ss use scalar product

•⁴ pd find scalar product

•⁵ pd find $|\overrightarrow{BA}|$

•⁶ ic find expression for $|\overrightarrow{BC}|$

•⁷ ic complete to result

•¹ $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$

•² $\begin{pmatrix} 2 \\ k+3 \\ -1 \end{pmatrix}$

•³ $\cos \hat{ABC} = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{|\overrightarrow{BA}| |\overrightarrow{BC}|}$ see Note 1

•⁴ 3

•⁵ $\sqrt{2}$

•⁶ $\sqrt{2^2 + (k+3)^2 + (-1)^2}$ or equivalent

•⁷ $\frac{3}{\sqrt{2}\sqrt{k^2 + 6k + 14}}$ and $\frac{3}{\sqrt{2(k^2 + 6k + 14)}}$

or $|\overrightarrow{BA}| |\overrightarrow{BC}| = \sqrt{2} \times \sqrt{k^2 + 6k + 14}$ and $\frac{3}{\sqrt{2(k^2 + 6k + 14)}}$

7

Notes

1. If the evidence for •³ does not appear explicitly, then •³ is only awarded if working for •⁷ is attempted.
2. •⁷ is dependent on gaining •⁴, •⁵ and •⁶.

Regularly occurring responses

Response 1 : Calculating wrong angle

Candidate A

$\cos AOC = \frac{\overrightarrow{OA} \cdot \overrightarrow{OC}}{|\overrightarrow{OA}| |\overrightarrow{OC}|}$ ✗ •³

$\overrightarrow{OA} \cdot \overrightarrow{OC} = 3 \times 4 + (-3) \times k + 0 \times 0 = 12 - 3k$ ✗ •⁴

$|\overrightarrow{OA}| = \sqrt{18}$ ✗ •⁵

$|\overrightarrow{OC}| = \sqrt{16 + k^2}$ ✗ •⁶

$\cos ABC = \frac{12 - 3k}{\sqrt{18}\sqrt{16 + k^2}}$ ✗ •⁷

Candidate B

$\cos AOB = \frac{\overrightarrow{OA} \cdot \overrightarrow{OB}}{|\overrightarrow{OA}| |\overrightarrow{OB}|}$ ✗ •³

$\overrightarrow{OA} \cdot \overrightarrow{OB} = 3 \times 2 + (-3) \times (-3) + 0 \times 1 = 15$ ✗ •⁴

$|\overrightarrow{OA}| = \sqrt{18}$ ✗ •⁵

$|\overrightarrow{OB}| = \sqrt{14}$ ✗ •⁶

$\cos ABC = \frac{15}{\sqrt{18}\sqrt{14}}$ ✗ •⁷

Generic Scheme

Illustrative Scheme

5(b)

Method 1 : Squaring first

- ⁸ ic link with (a)
- ⁹ ss square both sides
- ¹⁰ pd rearrange into 'non-fractional' format
- ¹¹ pd write in standard form
- ¹² pd solve for k

Method 2 : Dealing with fractions first

- ⁸ ic link with (a)
- ⁹ pd rearrange into 'non-fractional' format
- ¹⁰ ss square both sides
- ¹¹ pd write in standard form
- ¹² pd solve for k

Method 1 : Squaring first

- ⁸ $\frac{3}{\sqrt{2(k^2 + 6k + 14)}} = \cos 30^\circ$
- ⁹ $\left(\frac{3}{\sqrt{2(k^2 + 6k + 14)}}\right)^2 = \left(\frac{\sqrt{3}}{2}\right)^2$
- ¹⁰ $k^2 + 6k + 14 = 6$ or equivalent
- ¹¹ $k^2 + 6k + 8 = 0$ or equivalent
- ¹² $k = -2$ or -4

= 0 must appear at this stage.

Method 2 : Dealing with fractions first

- ⁸ $\frac{3}{\sqrt{2(k^2 + 6k + 14)}} = \cos 30^\circ$
- ⁹ $\sqrt{3}\sqrt{2(k^2 + 6k + 14)} = 6$
- ¹⁰ $6(k^2 + 6k + 14) = 36$
- ¹¹ $k^2 + 6k + 8 = 0$ or equivalent
- ¹² $k = -2$ or -4

= 0 must appear at this stage.

5

Notes

- The evidence for •⁹ may appear in the working for •¹⁰ in both methods.
- ⁹ is the only mark available to candidates who replace $\cos 30^\circ$ by 30 in method 1 and •¹⁰ in method 2.
- All 5 marks are available to candidates who use 0.87 for $\cos 30^\circ$ but 0.9 can gain a maximum of 4 marks.

Regularly occurring responses

Response 2 : Working with $\cos 30^\circ$ throughout the question

Candidate C (Method 1)

$$\begin{aligned}\cos 30^\circ &= \frac{3}{\sqrt{2(k^2 + 6k + 14)}} \quad \checkmark \bullet^8 \\ (\cos 30^\circ)^2 &= \left(\frac{3}{\sqrt{2(k^2 + 6k + 14)}}\right)^2 \quad \checkmark \bullet^9 \\ (\cos 30^\circ)^2 &= \frac{9}{2(k^2 + 6k + 14)} \\ 2(\cos 30^\circ)^2(k^2 + 6k + 14) &= 9 \quad \checkmark \bullet^{10}\end{aligned}$$

If $\cos 30^\circ$ is subsequently evaluated then •¹¹ and •¹² may still be available.Response 3 : Using the wrong value for $\cos 30^\circ$

Candidate D (Method 2)

$$\begin{aligned}\frac{3}{\sqrt{2(k^2 + 6k + 14)}} &= \frac{1}{2} \quad \times \bullet^8 \\ \sqrt{2(k^2 + 6k + 14)} &= 6 \quad \times \bullet^9 \\ 2(k^2 + 6k + 14) &= 36 \quad \times \bullet^{10} \\ k^2 + 6k + 14 &= 18 \\ k^2 + 6k - 4 &= 0 \quad \times \bullet^{11} \\ k &= \frac{-6 \pm \sqrt{6^2 - 4 \times 1 \times (-4)}}{2 \times 1} \\ &= 0.61, -6.61 \quad \times \bullet^{12}\end{aligned}$$

6 For $0 < x < \frac{\pi}{2}$, sequences can be generated using the recurrence relation

$$u_{n+1} = (\sin x)u_n + \cos 2x, \text{ with } u_0 = 1.$$

(a) Why do these sequences have a limit?

2

Generic Scheme

Illustrative Scheme

6 (a)

- | | |
|---|--|
| • ¹ ic condition on u_n coefficient | • ¹ $-1 < \sin x < 1$ |
| • ² ic connect coefficient with given interval | • ² in interval, $0 < \sin x < 1$ |

2

Notes

- For •¹ **do not** accept:
 - $\sin x$ lies between -1 and 1
 - $-1 < x < 1$
 - $-1 < \sin < 1$
 However, accept ' $\sin x$ greater than -1 and less than 1 ' for •¹.
- Do not accept $-1 < a < 1$ for •¹ unless a is clearly identified as $\sin x$, which may not appear until (b).
- $0 < \sin x < 1$ and nothing else, does not gain •¹ but gains •².
- $0 \leq \sin x \leq 1$ and nothing else, does not gain •¹ or •².

Regularly occurring responses

Response 1 : Attempts at giving a reason for limit

Candidate A

This sequence has a limit because $-1 < a < 1$,
 $-1 < \sin x < 1$ within the domain. ✓

•¹ ✓
 •² ✗

Candidate B

✓ Since $\sin x$ in this domain will always
 be greater than 0 and less than 1 . ✓

•¹ ✗
 •² ✓

Candidate C

$\sin \frac{\pi}{2} = 1$ and $\sin 0 = 0$ so the multiplier
 of u_n is between 0 and 1 , so it has a limit. ✓ ✓

•¹ ✗
 •² ✗

Candidate D

$-1 \leq \sin x \leq 1$,
 for $0 < x < \frac{\pi}{2}$, $0 < \sin x < 1$ ✓
 so limit exists ✓

•¹ ✗
 •² ✓

Response 2 : Minimum response for both marks

Candidate E

for $0 < x < \frac{\pi}{2}$, $0 < \sin x < 1$ •² ✓
 so $-1 < \sin x < 1$ •¹ ✓
 so limit

Candidate F

if limit, $-1 < \sin x < 1$ •¹ ✓
 for $0 < x < \frac{\pi}{2}$, $0 < \sin x < 1$ •² ✓

- 6 (b) The limit of one particular sequence generated by this recurrence relation is $\frac{1}{2}\sin x$.
Find the value(s) of x .

7

Generic Scheme

Illustrative Scheme

6 (b)

•³ ss appropriate limit method

•⁴ ic substitute for limit

•⁵ ss use appropriate double angle formula

•⁶ pd express in standard form

•⁷ pd start to solve quadratic equation

•⁸ pd reduce to equations in $\sin x$ only

•⁹ ic select valid solution

•³ limit = $\frac{\cos 2x}{1 - \sin x}$ or $l = \sin x \times l + \cos 2x$

•⁴ $\frac{1}{2}\sin x = \frac{\cos 2x}{1 - \sin x}$ or $\frac{1}{2}\sin x = \sin x \times \frac{1}{2}\sin x + \cos 2x$

(•³ may be stated, or implied by •⁴ in both methods)

•⁵ $\dots 1 - 2\sin^2 x \dots$

•⁶ e.g. $3\sin^2 x + \sin x - 2$ } = 0 must appear at •⁶ or •⁷

•⁷ e.g. $(3\sin x - 2)(\sin x + 1)$ } to gain •⁶.

•⁸ $\sin x = \frac{2}{3}$ or $\sin x = -1$

•⁹ $x = 0.730$ or outwith interval

7

Notes

- ⁷, •⁸ and •⁹ are only available if a quadratic equation is obtained at •⁶ stage.
- Candidates may express the quadratic equation at the •⁶ stage in the form $3s^2 + s - 2 = 0$. For candidates who do not solve a trigonometric quadratic equation at •⁷ $\sin x$ must appear explicitly to gain •⁸.
- ⁷, •⁸ and •⁹ are not available to candidates who 'solve' a quadratic equation in the form $ax^2 + bx = c$, $c \neq 0$.
- For •⁹ there must be one valid solution, and one solution outwith interval which is rejected.
- ⁹ is not available to candidates who leave their answer in degree measure.
- Cross marking is available for •⁸ and •⁹.

Regularly occurring responses

Response 3 : Evidence for identification of a appearing in (b)

Candidate G

(a) $-1 < a < 1$ ✓

(b) $L = \frac{b}{1-a} = \frac{\cos 2x}{1-\sin x}$ ✓ •³

•¹ ✓

•² ✗

•³ ✓

Response 4 : Error in algebra and subsequent quadratic equation solution

Candidate H

$L = \frac{b}{1-a} = \frac{1}{2}\sin x$

$\frac{\cos 2x}{1-\sin x} = \frac{1}{2}\sin x$ ✓ •³ ✓ •⁴

$\cos 2x = -\frac{1}{2}\sin^2 x$ ✗ •⁶

$\frac{1}{2}\sin^2 x + \cos 2x = 0$

$\frac{1}{2}\sin^2 x + (1 - 2\sin^2 x) = 0$ ✗ •⁵

$-\frac{3}{2}\sin^2 x + 1 = 0$

$\sin^2 x = \frac{2}{3}$ ✗ •⁷

$\sin x = \sqrt{\frac{2}{3}}$ and $\sin x = -\sqrt{\frac{2}{3}}$ ✗ •⁸

$x = 0.955, 2.186$ ~~$x = 4.097, 5.328$~~ ✗ •⁹

Candidate I

$\frac{\cos 2x}{1-\sin x} = \frac{1}{2}\sin x$ ✓ •³ ✓ •⁴

$\frac{1}{2}\sin x(1-\sin x) = 1 - \sin^2 x$ ✗ •⁵

$\sin^2 x + \sin x - 2 = 0$ ✗ •⁶

$(\sin x - 1)(\sin x + 2) = 0$ ✗ •⁷

$\sin x = 1$ and $\sin x = -2$ ✗ •⁸

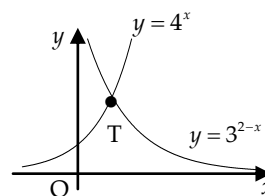
$x = \frac{\pi}{2}$ not possible ✗ •⁹

See Note 8

- 7 The diagram shows the curves with equations $y = 4^x$ and $y = 3^{2-x}$.

The graphs intersect at the point T.

- (a) Show that the x -coordinate of T can be written in the form $\frac{\log_a p}{\log_a q}$, for all $a > 1$.



6

Generic Scheme

Illustrative Scheme

7(a)

- ¹ ss equate expressions for y
- ² ss take logarithms of both sides
- ³ ic use law of logs: $\log_a x^n = n \log_a x$
- ⁴ pd gather like terms
- ⁵ ic use law of logs: $\log_a p + \log_a q = \log_a pq$
- ⁶ ic complete to required form

In methods 1 and 2:

If the first line of working is that at the •² stage, then •¹ and •² are awarded.

If the first line of working is that at the •³ stage, then only •² and •³ are awarded.

Method 1

- ¹ $4^x = 3^{2-x}$
- ² $\log_a (4^x) = \log_a (3^{2-x})$ **stated, or implied by** •³
- ³ $x \log_a 4 = (2-x) \log_a 3$
- ⁴ $x(\log_a 4 + \log_a 3) = 2 \log_a 3$
- ⁵ $x \log_a 12 = \log_a 9$
- ⁶ $\frac{\log_a 9}{\log_a 12}$ **stated explicitly**

Method 2

- ¹ $4^x = 3^{2-x}$
- ² $\log_3 (4^x) = 2-x$
- ³ $x \log_3 4 = 2-x$
- ⁴ $x = \frac{2}{1 + \log_3 4}$
- ⁵ $\frac{2 \log_3 3}{\log_3 12}$
- ⁶ $\frac{\log_a 9}{\log_a 12}$ **stated explicitly**

Method 3

- ¹ $4^x = 3^{2-x}$
- ² $4^x = \frac{3^2}{3^x}$
- ³ $12^x = 9$
- ⁴ $\log_a 12^x = \log_a 9$
- ⁵ $x \log_a 12 = \log_a 9$
- ⁶ $\frac{\log_a 9}{\log_a 12}$ **stated explicitly**

6

Notes

- In methods 1 and 2, if no base is indicated then •² is not available, however •³, •⁴ and •⁵ are still available. In method 3, if no base is indicated then •⁴ is not available, however •⁵ is still available.
- In all methods, if a numerical base is used then •⁶ is not available.
- In method 1, the omission of brackets at the •³ stage is treated as bad form, see Response 1.
- p and q must be numerical values.

Regularly occurring responses

Response 1: Omission of brackets around $2-x$

Candidate A $4^x = 3^{2-x}$ ✓ •¹
 $x \log_a 4 = 2 - x \log_a 3$ ✓ •² ✓ •³

Candidate B

$$4^x = 3^{2-x} \quad \checkmark \bullet^1$$

$$x \log_a 4 = 2 - x \log_a 3 \quad \checkmark \bullet^2 \quad \checkmark \bullet^3$$

$$\underline{x(\log_a 4 + \log_a 3) = 2} \quad \times \bullet^4$$

$$x \log_a 12 = 2 \quad \times \bullet^5$$

Response 2: Using different bases
Candidate C

$$4^x = 3^{2-x} \quad \checkmark \bullet^1$$

$$\log_3 4^x = \log_4 3^{2-x} \quad \times \bullet^2$$

$$x \log_3 4 = (2-x) \log_4 3 \quad \times \bullet^3$$

Response 3: Taking logs first
Candidate D

$$y = 4^x \text{ and } y = 3^{2-x}$$

$$\log_a y = \log_a 4^x \text{ and } \log_a y = \log_a 3^{2-x} \quad \checkmark \bullet^2$$

$$\log_a y = x \log_a 4 \text{ and } \log_a y = (2-x) \log_a 3 \quad \checkmark \bullet^3$$

$$x \log_a 4 = (2-x) \log_a 3 \quad \checkmark \bullet^1$$

$$x = \frac{2}{\log_a 12}$$

$$= \frac{2 \log_a a}{\log_a 12}$$

$$= \frac{\log_a a^2}{\log_a 12} \quad \times \bullet^6$$

Generic Scheme

Illustrative Scheme

7(b)

- ⁷ ic substitute in for x
- ⁸ pd process y

- ⁷ e.g. $y = 4^{\frac{\log_a 9}{\log_a 12}}$
- ⁸ e.g. $y \approx 4^{0.8842} \approx 3 \cdot 4$

stated, or implied by •⁸

2

Notes

5. Candidates must work to at least two significant figures in (b) e.g. $4^{0.9} = 3.5$ does not gain •⁸, but •⁷ is available.
6. •⁸ is only available if the power used comes from $\frac{\log_a p}{\log_a q}$ in (a).

Regularly occurring responses

Response 4 : Using p and q as integer values without working

Candidate E

$$\left. \begin{array}{l} p = 4 \\ q = 3 \end{array} \right\} y = 4^{1.26} = 5.74 \text{ or } 5.75$$

✗ •⁷
✗ •⁸

Candidate F

$$\left. \begin{array}{l} p = 3 \\ q = 4 \end{array} \right\} y = 4^{0.79} = 2.99 \text{ or } 3$$

✗ •⁷
✗ •⁸

Response 5 : Using integer values calculated in (a)

Candidate G

$$\left. \begin{array}{l} p = 10 \\ q = 4 \end{array} \right\} y = 4^{2.5} = 32$$

✗ •⁷
✗ •⁸

[END OF MARKING INSTRUCTIONS]